snowflake®

Cooking with CoCo

Stop Prompting. Start Specifying.

A Spec-Driven Approach to AI-Assisted Data Engineering



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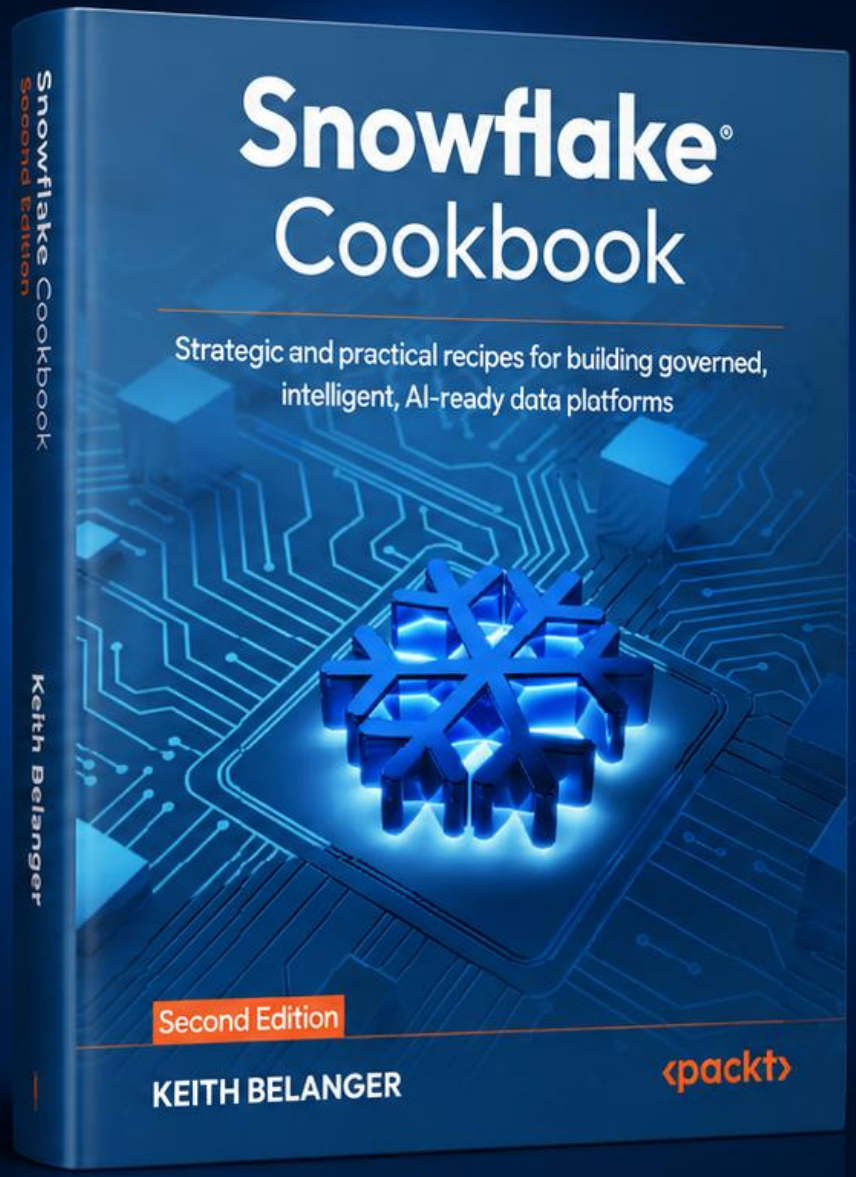
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ABOUT YOUR HOST

Keith Belanger

DATA ARCHITECT | AUTHOR | SPEAKER

- 30+ years designing and delivering enterprise data solutions that drive business outcomes.
- Expert in data architecture, data modeling (3NF, Kimball, Data Vault 2.0), and data governance.
- Snowflake Data Superhero and passionate advocate for the data community.
- Frequent speaker, author, and mentor—helping organizations build trusted, AI-ready data platforms.
- Bridging strategy, modeling, and engineering in the age of AI.



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Are We Vibe Coding Our Data Platforms?

The problem with prompting without a specification.



We describe what we want.
We prompt the AI with our intent and desired outcome.

AI generates something plausible.
The output looks reasonable, compiles, and runs.

The human becomes the test suite.
We review it, spot check it, and decide if it's "good enough."

But how does the AI know what "right" means?

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The Spec-Driven Approach

Iterate on the Specification, not the prompt.

We define the outcome, context, and standards up front.
AI engineers within those guardrails to deliver the right solution.

1. SPECIFY
Define Success

- Business Goals
- Source Context
- Governance & Standards
- Acceptance Criteria



This is our contract for success.

2. AI ENGINEERS
Within Guardrails



3. IMPLEMENT
AI Builds the Solution

- Design ✓
- Code ✓
- Tests ✓
- Documentation ✓
- Validation ✓

4. VALIDATE
Against the Spec



5. REFINE THE SPEC
If Needed



Update the specification if the outcome needs to change.

We focus on the "why", not the "how". | AI executes within the context and standards.

Clear Outcomes
Defined and agreed before implementation.

Consistent Results
AI follows the same standards every time.

Higher Quality
Validated against a shared contract.

Faster Alignment
Less debate, more delivery.

Scalable Success
Reusable specs, context, and standards.

The Three Foundations

Organizational knowledge that gives AI the context to engineer the right solution.

1. GOVERNANCE

Technology-agnostic organizational policies and standards.

Examples:

- Business terminology
- Data quality expectations
- Modeling principles
- Naming philosophy
- Security / governance expectations
- Semantic consistency

Defines how we work, what we mean, and what good looks like.

2. ARCHITECTURE & ENGINEERING STANDARDS

Technology-specific implementation standards.

Examples:

- Snowflake engineering
- dbt engineering
- Semantic modeling
- Horizon governance implementation
- Platform-specific patterns

Defines how we implement, engineer, and operate on our chosen platform.

3. SOURCE CONTEXT

What the physical source data actually means.

Examples:

- Source systems
- Tables and columns
- Relationships
- Status semantics
- Source-specific interpretation
- Mapping physical structures to business concepts

Defines the real-world meaning behind the data structures we work with.

Your organization needs these whether the engineer is human or AI.

Consistency
Same language, same meaning, same quality.

Alignment
Everyone builds toward the same outcomes.

Governance
Security, compliance, and trust by design.

Reuse & Scale
Codified knowledge that scales across teams.

AI doesn't create the need for these foundations. It exposes the risk when they only exist in people's heads.

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Why AI Changes the Equation

Same foundations. New capability.

For thirty years we've talked about standards, governance, architecture and metadata.

The difference now: these artifacts **aren't just documentation for people. They can become executable context for AI.**

> **If organizational knowledge isn't explicit, the AI has to guess.**

HISTORICALLY

Organizational Knowledge
- Governance
- Standards
- Source Context
- Best Practices

→ **HUMAN ENGINEERS**
Interpret, decide, and build.

→ **IMPLEMENTATION**
Solutions built and deployed.

NOW

Organizational Knowledge
- Governance
- Standards
- Source Context
- Best Practices

→ **HUMAN ENGINEERS**
Provide direction, ask the right questions, validate outcomes.

\+ **AI ENGINEERS**
Ingest the context, make decisions, and build.

→ **IMPLEMENTATION**
Solutions built and deployed.

Explicit context leads to consistent, trustworthy, and higher-quality outcomes.

- **Less Guessing** — More Confidence
- **Aligned Outcomes** — Every Time
- **Higher Quality** — Lower Rework
- **Scalable Knowledge** — Across Teams

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The Business Specification

Define the outcome. Provide the context. Let AI engineer the solution.

THE FOURTH PIECE

The three foundations describe how the organization operates and what the environment means.

The Business Specification describes:

What are we actually trying to accomplish?

NOTICE WHAT ISN'T HERE.

- ⊗ No TPCH table names
- ⊗ No dbt models
- ⊗ No star schema design
- ⊗ Not prescribing the implementation

✓ Defining the business outcome, not the implementation.

EXAMPLE: SALES ANALYTICS SPECIFICATION

1. BUSINESS OBJECTIVE	Provide a trusted, self-service sales analytics solution that enables leadership to monitor performance, understand trends, and drive data-informed decisions.
2. TERMINOLOGY	Define key business terms and their definitions to ensure consistent understanding. Examples: Customer, Order, Sales, Revenue, Margin, Product, Territory, Channel
3. KPIs	Define the key performance indicators that measure success. Examples: Total Revenue, Gross Margin %, Orders, AOV, New Customers, Churn Rate
4. BUSINESS QUESTIONS	Define the key questions the solution must answer. Examples: How are we performing vs plan? Which products are driving growth? Which customers are at risk? What is the revenue by segment over time?
5. REQUIRED CAPABILITIES	Define the capabilities the solution must provide. Examples: Drill to detail, Time comparisons, Segmentation, Export, Alerts, Row-level security, Data freshness
6. ACCEPTANCE CRITERIA	Define how we know the solution is successful. Examples: KPI values match source of record, Performance SLA met, Security requirements validated, User acceptance validated

SOURCE CONTEXT + GOVERNANCE + ARCHITECTURE & STANDARDS + BUSINESS SPECIFICATION → **Now we have the outcome, the context, and the organizational standards.**

This sets the AI engineer up with everything it needs to design and build the right solution.

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DEMO

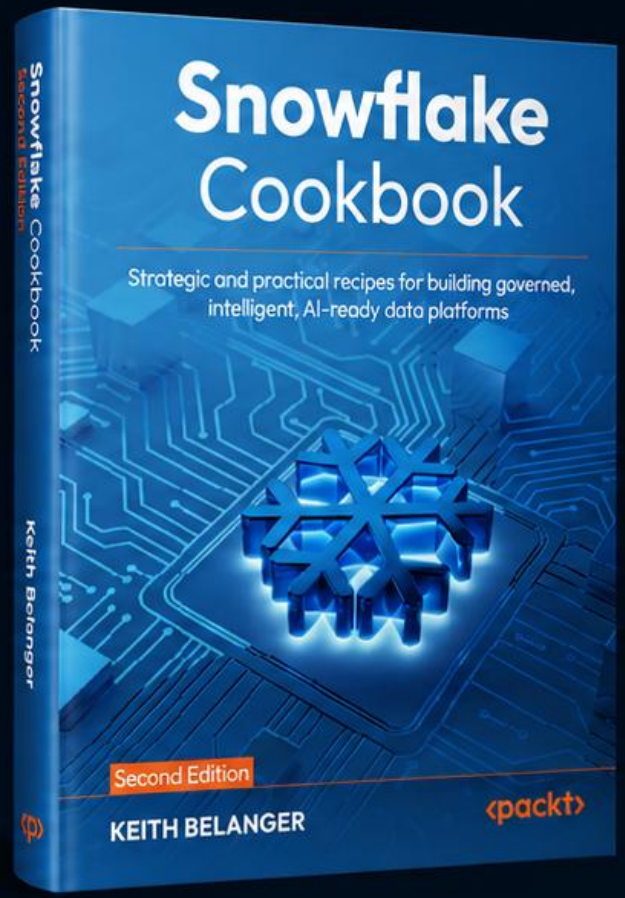
Let's see it in action.

Real Data | AI Engineer | Live Build | Real Results

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Thank You!

Want to go deeper?
Get the complete guide.



Available on amazon

Scan to purchase now!

Stop prompting. Start specifying.

Good architecture, governance, and engineering standards matter more in the age of AI—not less.

THE FUTURE BELONGS TO THOSE WHO SPECIFY IT.

- **Define** the outcome.
- **Provide** the context.
- **Apply** the standards.
- **Let AI** engineer.
- **Validate** the result.

That's how we build trusted, **AI-ready** data platforms.

- **100+ practical recipes** for real-world Snowflake implementations
- **From foundational** concepts to advanced architectures
- **Best practices,** patterns, and proven solutions
- **Written for data** professionals by a practitioner



How Was I?